AI Tools for Learning & Education

How to use LLMs and AI tools to learn better, not less

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The AI Education Explosion

85% of teachers and **86%** of students

used AI during the 2024–2025 school year

All three major LLMs launched education-specific modes within weeks of each other:

ChatGPT	**Claude**	**Gemini**
Study Mode	Learning Mode	Guided Learning
July 2025	August 2025	August 2025

The question isn't *whether* to use AI — it's **how to use it well**.

The Core Tension

AI can give you any answer instantly

Copy question → paste answer
Zero effort, zero learning

your choice →

AI can help you understand deeply

Guide, quiz, explain, challenge
Real learning, lasting knowledge

"AI should make students think harder, not less." — Anthropic

What We'll Cover

1. **The Big Three**
   - ChatGPT Study Mode
   - Claude Learning Mode
   - Gemini Guided Learning
   - ArmBench-LLM
2. **Specialized Tools**
   - NotebookLM
   - Khanmigo
   - Language learning tools
3. **How to Learn**
   - The right way vs. wrong way
   - Prompting for learning
   - Scheduling, voice, custom instructions
4. **Pitfalls & Tips**
   - Hallucinations
   - Free access guide
   - Resources & links

ChatGPT Study Mode

Launched: July 2025 | **Access:** Free tier (select "study and learn" in tools)

How it works:

- Guides you step-by-step instead of giving answers
- **Refuses** direct answers until you engage with the material
- Asks questions to test your understanding
- Calibrates to your skill level

Interactive Learning:

- 70+ math & science topics
- Pythagorean theorem, ideal gas law, lens equations...
- Visualizations for complex concepts

Example interaction

You: How do I find the derivative of $f(x) = x^3 + 2x$?

ChatGPT: Before I show you, let me ask -- do you remember the power rule? What happens when you differentiate x^n?

You: nx^{n-1}?

ChatGPT: Exactly! Now try applying that to each term...

ChatGPT Study Mode: Features

Upcoming features

- Clearer visualizations for complex concepts
- Goal setting & progress tracking
- Deeper personalization by skill level

Also useful for learning:

- **Canvas** — side-by-side editing for writing/code
- **Memory** — remembers your course, level, preferences
- **Custom GPTs** — course-specific chatbots
- **Voice mode** — conversational tutoring
- **File upload** — problem sets, diagrams

Demo idea

Open ChatGPT → "study and learn" → try a math problem. Notice how it guides rather than solves.

ChatGPT Edu:

- Dedicated tier for universities
- GPT-4o access, enterprise-grade privacy
- Admin controls, SSO, group workspaces

Claude: Learning & Explanatory Modes

Launched: August 2025 | **Stat:** 76% of users reported improved understanding

Learning Mode

Socratic pair-programming

- Pauses and inserts `#TODO` for you to write code
- You complete 5–10 lines yourself
- Claude reviews your work and guides further
- Learn by doing, not by reading

Explanatory Mode

Senior dev mentoring

- Narrates every decision as it works
- Explains architecture choices, tradeoffs
- Shares best practices in real-time
- Like looking "under the hood"

Claude for Education (since April 2025): deployed at Northeastern (50K users), LSE, Champlain College. *"AI should make students think harder, not less."*

Claude: Example Interaction

Learning Mode example

```
You:  Help me write a function
that reverses a linked list.

Claude:  Let's build this together.
I'll set up the structure --
you write the core logic.

def reverse_list(head):
  prev = None
  current = head
  # TODO: Write a while loop that
  # iterates through the list and
  # reverses the pointers.
  # (about 5 lines)
  return prev
```

Explanatory Mode example

```
You:  Explain binary search.

Claude:  Binary search works on
sorted arrays.  The key insight
is:  by checking the middle element,
we eliminate half the remaining
candidates each step.

Why sorted?  Because if arr[mid]
is too large, everything to its
right is also too large.  This
guarantee is what makes the
O(log n) complexity possible.

Think of it like a phone book...
```

Demo: Ask Claude to teach you a concept in Learning or Explanatory mode

Gemini Guided Learning

Launched: August 2025 | **Built on:** LearnLM — Google's education-specific AI

How it works:

- Socratic method + step-by-step guidance
- Won't just give you answers
- Asks questions to ensure you learned the concept
- LearnLM integrated into Gemini 2.5

Key features:

- **Canvas quizzes** — personalized quizzes for any subject
- **Video Overviews** — turn sources into educational videos
- **SAT prep** — free practice tests (Princeton Review)
- Free for educational institutions

LearnLM stats

- Only **0.1%** factual error rate
- Students **5.5 pp** more likely to solve novel problems
- Preferred by educators over competitors

Demo idea

Open Gemini → Guided Learning → ask about a concept. Notice the Socratic style.

Gemini: More Education Features

Gemini for Students:

- Integration with Google Classroom, Docs, Slides
- Ask questions while reading documents
- Get help with assignments
- Research topics within your workflow

For teachers:

- Draft assignments in Classroom
- Summarize student progress
- Record/attach audio & video for feedback

Example interaction

```
You: What causes ocean tides?

Gemini: Good question! Before I
explain, let me check what you
already know. Can you think of
what celestial body might have the
biggest influence on tides?

You: The moon?

Gemini: Right! Now, why do you
think the moon specifically?
What property does it have that
could affect Earth's oceans?
```

Side-by-Side Comparison

	ChatGPT	Claude	Gemini
Mode name	Study Mode	Learning / Explanatory	Guided Learning
Approach	Refuses answers, quizzes	Socratic + `#TODO` inserts	Socratic questioning
Best for	Math/science, general study	Coding, technical concepts	Google ecosystem, quizzes
Unique	70+ interactive topics	Pair-programming style	LearnLM, Video Overviews
Free tier	Yes	Yes	Yes (free for edu)
Ecosystem	Canvas, Custom GPTs	Artifacts, Projects	Classroom, Docs, Slides

Key insight: all three are designed to **guide**, not **give answers**.
Try the same question in all three — you'll notice different tutoring styles.

ArmBench-LLM: Which Models Speak Armenian?

ArmBench-LLM 1.0 — first comprehensive benchmark for Armenian language tasks. Created by **Metric AI Lab** (April 2026). Fully open-sourced.

7 task categories: text classification, multiple choice QA, grammar correction, space fixing (OCR), summarization, translation, reading comprehension.

Rank	Model	Score	Cost
1	**Gemini 3 Flash**	0.635	$3.28
2	GPT-5.2 Pro	0.617	$160.20
3	Grok 4 Fast	0.604	$2.71
4	Claude 3.7 Sonnet	0.607	$16.49
5	**Qwen 3.5 27B** (open-source)	0.577	free

Key insight

Global rankings $\neq$ Armenian proficiency! Gemini Flash **beats** Gemini Pro on Armenian (opposite of global ranking).

Open-source win

Qwen 3.5 27B beats 600B+ models (GLM-5, Mistral-Large). Runs on a **single GPU**.

ArmBench: Task-Specific Results

Best models by task:

Task	Best model
Translation	Gemini 3 Flash
Summarization	Gemini 3 Flash
Armenian Literature	Gemini 3 Flash
Armenian History	Gemini 3 Pro
Math (university exams)	Grok 4 Fast
Named Entity Recog.	Grok 4 Fast

Cost matters:

- GPT-5.2 Pro costs **50**× more than Gemini Flash for *lower* accuracy
- Armenian tokenizers vary wildly — same text = different cost
- Reasoning models burn tokens on chain-of-thought

Practical recommendations:

Best overall

Gemini 3 Flash — highest score, lowest cost.
Best for translation, summarization, literature.

Best budget

Grok 4 Fast — nearly as good, cheapest ($2.71).
Best for math, NER.

Best open-source

Qwen 3.5 27B — run locally, no API costs, beats much larger models.

Try it yourself

Leaderboard: `metric-ai-armbench-llm.hf.space`
Search `Metric-AI/ArmBench-LLM-data` on HF

NotebookLM: Your Personal Study Assistant

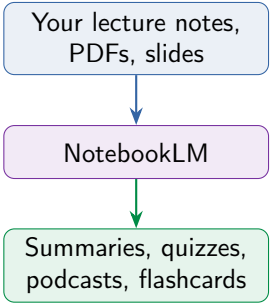
What: Upload YOUR materials → AI assistant grounded in YOUR sources

How it's different from chatbots:

- Regular LLMs answer from training data (generic internet)
- NotebookLM answers from **your uploaded documents**
- Source-grounded = fewer hallucinations
- Every answer references your materials

What you can upload:

- PDFs, Google Docs, Slides
- Lecture notes, textbook chapters
- Research papers, YouTube videos, audio



NotebookLM: Key Features

Audio Overviews

AI generates a **podcast-style discussion** of your materials. Two AI hosts discuss your content. Listen on commute, exercising, etc.

Interactive Mode

"Raise your hand" during the podcast. AI hosts stop, answer your question, then continue.

Flashcards & Quizzes

Auto-generated from your documents. Test yourself on **your actual course content**.

Learning Guide

Structured, deeper exploration of your materials. Beyond summaries — builds understanding.

Price

Free for all education users.

Demo: upload a lecture PDF, generate an Audio Overview, try Interactive Mode

Khanmigo: Purpose-Built AI Tutor

What: Khan Academy's AI tutor — designed for education from day one

How it works:

- Socratic method: guides, **never gives direct answers**
- Limitless patience — guides until you get it
- Built on GPT-4, constrained for education
- Integrated with Khan Academy's content

Features:

- **Image upload** — scan handwritten math
- Writing coach, debate partner
- Career coaching, speech support

Availability

- $4/month or $44/year
- Free via many school districts
- 10+ languages: English, Spanish, Russian, Arabic, Hindi, Chinese...

For teachers:

- Standards-aligned lesson plans
- Student progress summaries
- Learning objectives, rubrics, quizzes

AI for Language Learning

Dedicated tools

- **Duolingo** — gamified, AI-adaptive, roleplay conversations
- **Speak** — AI conversation partner
- **ELSA Speak** — pronunciation feedback

Why LLMs are game-changers:

- Free-form conversation in any language
- Correct mistakes with explanations
- Adapt to your level automatically
- Available 24/7

Using LLMs for languages

- **Voice modes** — speak and listen
- Practice real conversations
- Grammar explanations in your language
- Role-play: ordering food, interviews...

Try this prompt

"Let's talk in French. I'm B1. Correct my mistakes gently and explain why."

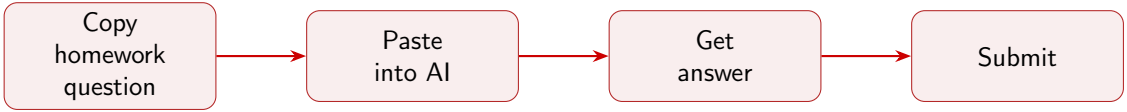
The AI Learning Toolkit

What do you need? → Use this tool

Task	Best Tool	Why
Study from your notes	NotebookLM	Grounded in your materials
Understand a concept	ChatGPT / Claude / Gemini	Learning/education modes
Guided tutoring	Khanmigo	Socratic method, never gives answers
Practice a language	Duolingo + LLM voice	Structured + free-form practice

Best results come from combining tools, not relying on one.

The Wrong Way



What happens

- Zero effort → zero learning
- False confidence on exams
- Fragile knowledge that crumbles under pressure
- Can't explain your own work

The exam problem

- You can't bring AI to the exam
- If you never struggled with the material, you never learned it
- **Struggle is where learning happens**

The Right Way: AI as Tutor, Not Calculator

1. Try first Attempt the problem → **2. Get stuck** Identify what you don't get → **3. Ask AI** to *guide*, not solve → **4. Do it** yourself with guidance → **5. Verify** Cross-check with textbook

The key difference

- "Help me understand X" ✓
- "Solve X for me" ×
- You do the thinking — AI guides
- Learning modes enforce this!

Why this works

- You still struggle (that's learning!)
- AI provides scaffolding, not answers
- You build real understanding
- Knowledge that survives exams

Prompting for Learning (Part 1)

Prompts that maximize learning:

Guide, don't solve

- *"Don't give me the answer. Guide me step by step."*
- *"I think the answer is X. Am I right? Where did I go wrong?"*
- *"What's wrong with my approach?"*

Explain & understand

- *"Explain this using an analogy."*
- *"What's the intuition behind [formula]?"*
- *"Why does this work? Not how — why."*

Test yourself

- *"Quiz me on [topic] — start easy, then harder."*
- *"Give me 5 practice problems similar to this."*
- *"I studied X. Check if I really understand."*

Go deeper

- *"What concept am I missing?"*
- *"What are common misconceptions here?"*

Prompting for Learning (Part 2)

More advanced techniques:

Level adjustment

- *"Explain like I'm a complete beginner."*
- *"Now explain at a graduate level."*
- *"Use only concepts from Calculus 1."*

Connections & synthesis

- *"How does [A] relate to [B]?"*
- *"Real-world example of [concept]."*
- *"Compare and contrast [X] and [Y]."*

Role-based learning

- *"Act as a [subject] professor — give me a 5-min lecture on [topic]."*
- *"Be my study partner. Let's challenge each other."*

Exam prep

- *"Study plan for [topic], 2h/day, 5 days."*
- *"Practice exam: 10 questions on ch. 5–8."*
- *"Most likely exam questions on [topic]?"*

Scheduled Prompts: Something New Every Day

Schedule a **daily recurring prompt** — the AI sends it automatically.
Micro-learning compounds — 5 minutes/day = 30 hours/year.

How to schedule in each tool

- **ChatGPT:** type *"Every morning at 9am, send me..."* — it creates a scheduled task automatically
- **Claude:** Projects → set a recurring prompt in project instructions
- **Gemini:** use Google Calendar + Gemini integration, or Gems with reminders

Example prompts to schedule

- *"Teach me one new statistics concept in 3 sentences."*
- *"Give me one Python trick I probably don't know."*
- *"One interesting fact about linear algebra."*
- *"Explain one ML algorithm in simple terms."*

Set it once, learn every day — zero effort after setup.

Custom Instructions: Set the Rules Once

All major LLMs let you set persistent instructions that apply to every conversation.

What are custom instructions?

- Rules the AI follows in **every** conversation
- Set once, applies automatically
- Control tone, depth, format, behavior

Where to find them

- **ChatGPT:** Settings → Custom Instructions
- **Claude:** Settings → Profile
- **Gemini:** Settings → Gems

Example instructions

```
I'm a 2nd-year CS student.
When I ask about code, don't give
me the answer directly -- guide me
step by step.
Use simple analogies.
Quiz me after explaining a concept.
If I ask in Armenian, respond in
Armenian.
```

Pro tip

Update your instructions as you progress. What works for a beginner won't work 6 months later.

Passive Language Learning in Every Chat

Learn vocabulary without extra effort — just by chatting about anything.

The idea

Ask the LLM to replace 1–2 common words in every response with translations in your target language. You absorb vocabulary **passively** while doing your normal work.

What it looks like

```
You:  Explain how a hash map works.

AI: A hash map stores Schlüssel (keys) and
their associated values.  When you insert
data, a Funktion (function) converts the key
into an index...
```

Example prompt

"In every response, replace 1–2 common words with their German translation (in parentheses)."

Power move

Put this in your **custom instructions** and it happens in every conversation automatically — zero extra effort.

Works for **any** language pair. Great for German, French, Spanish, or even Armenian!

Voice Mode: Practice Job Interviews with AI

Use voice mode to simulate real interviews — out loud, in real time.

How it works

1. Find a job posting on LinkedIn
2. Paste it into ChatGPT / Claude / Gemini
3. Say: *"Interview me for this role. Ask me technical and behavioral questions one at a time."*
4. Switch to **voice mode** and answer out loud
5. Ask for feedback after each answer

Example prompt

```
Here's a job posting for a Junior
Data Analyst at [company]:

[paste the LinkedIn post]

Interview me for this position.
Ask one question at a time.  After
I answer, give brief feedback and
then ask the next question.  Start
with a behavioral question, then
move to technical.
```

Why voice matters

- Real interviews are **spoken**, not typed
- Practice articulating under pressure
- Builds confidence before the real thing

Creative LLM Learning Hacks

Unconventional ways to squeeze more learning from every interaction:

Passive language immersion

"Replace 3–5 words in every response with their French equivalent (in parentheses)." Absorb vocabulary **without dedicated study time**.

Daily drip — scheduled learning

Set a daily reminder to prompt the LLM: *"Teach me one new concept about [topic] in 3 sentences."* Micro-learning compounds over weeks.

"Always teach me something new"

"End every response with one surprising fact about what we discussed." Turns every chat into discovery.

Teach-back method

Explain a concept **to the LLM**, then ask: *"Find the holes in my explanation."* Teaching is the best way to learn.

"Debug my thinking"

"Here's my mental model of [X]. Where am I wrong?" Surfaces blind spots you didn't know you had.

Counterfactual exploration

"What if Newton never discovered calculus?" "What if" questions build deeper intuition by exploring **why** things are the way they are.

Combining Tools: A Study Session Workflow

1. Upload lecture notes to **NotebookLM** → listen to Audio Overview on commute
2. Ask **ChatGPT/Claude/Gemini** in learning mode to quiz you
3. Use **Khanmigo** for guided practice problems
4. Create summary flashcards in **NotebookLM** for spaced review

Each tool has a strength — combine them for maximum learning.

Hallucinations: AI Makes Stuff Up

What are hallucinations?

- LLMs generate plausible-sounding but **wrong** info
- They don't "know" facts — they predict likely text
- Can invent citations, formulas, historical events
- Sounds confident even when completely wrong

Educational risks:

- Learning wrong formulas or definitions
- Citing non-existent papers
- Building understanding on incorrect foundations

How to protect yourself

1. **Always verify** with textbook/professor
2. Use **NotebookLM** for source-grounded answers
3. **Cross-reference** multiple tools
4. Be skeptical of confident answers

Rule of thumb

If AI says something surprising, verify before relying on it.

Key Takeaways

1. Use **learning/education modes**, not default modes
2. AI should make you **think more**, not less
3. **Combine tools** — each has different strengths
4. **Always verify** — AI hallucinates
5. **Be transparent** about AI use

Free Access for Students

Tool	Free access	How
ChatGPT	Study Mode on free tier	Sign up at chat.openai.com
Claude	Free tier available	Sign up at claude.ai
Gemini	Free for edu institutions	Via Google Workspace for Education
NotebookLM	Free for everyone	notebooklm.google
Khanmigo	$4/mo (or free via school)	khanmigo.ai
Duolingo	Free tier available	duolingo.com

Most of these tools are free or have generous student programs.
You have no excuse not to try them!

Resources & Links

The Big Three:

- ChatGPT: `chat.openai.com`
- Claude: `claude.ai`
- Gemini: `gemini.google.com`

Specialized Tools:

- NotebookLM: `notebooklm.google`
- Khanmigo: `khanmigo.ai`
- Duolingo: `duolingo.com`

Questions?

Use AI to learn better — not to learn less.